PRANAV SREERAM

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EDUCATION

Carnegie Mellon University **Pittsburgh, PA**
M.S, Machine Learning *Expected Graduation December 2027*

- **Relevant Courses:** Introduction to Computer Systems, Deep Learning Systems, Probability & Mathematical Statistics, Advanced Topics in Machine Learning

University of California, Los Angeles **Los Angeles, CA**
B.S, Mathematics/Economics *Graduated June 2026*

- **Academics and Activities:** GPA: 3.9/4.0, 5x Dean's Honor List, Co-Founder of Bruin Quant Traders
- **Relevant Courses:** Algorithms, Artificial Intelligence, Cryptography I & II (Graduate), Data Structures, Deep Learning I & II, Linear Algebra, Machine Learning, Optimization**,** Probability, Statistics, Stochastic Processes

WORK EXPERIENCE

Blackrock **New York, NY**
Software Engineering Intern *June 2026 - August 2026*

- Spearheaded development of AI tooling for Aladdin Research that transforms fragmented analyst research notes and investment opinions into structured intelligence, accelerating synthesis of insights for 170 active clients/teams
- Architected a multi-agent AI research pipeline in Python using async orchestration, an agent-to-agent protocol, and parallel task dispatch to synthesize investment insights from three source research APIs, so far achieving 88% grounding accuracy across a test set of 497 insights on complex queries about various companies and sectors
- Developed an LLM-as-judge evaluation framework with role-aware prompt tuning and human-label calibration (Cohen's κ), reducing manual insight review from hours of analyst time to automated single-command assessment
- Engineered a reproducible ablation infrastructure with frozen corpus snapshots, bulk evaluation runners, and A/B statistical comparison to systematically benchmark 28 pipeline variants, enabling data-driven optimizations

Chicago Trading Company **Chicago, IL**
Quantitative Trading Intern *June 2025 - August 2025*

- Trained in options theory, market making, and quant modeling, competing against other interns in using mental math, statistics, and data science to price, quote, and trade S&P 500 options in a simulated exchange pit
- Investigated trading flows in the gold options market, testing for signals to predict edge, volume, and trading activity using rigorous statistical tests and finding significant relationships in historic options data
- Leveraged NumPy, SciPy, and SQL to generate regression models and perform hypothesis tests, querying millions of trades from Snowflake tables and developing a pipeline to clean data, generate features, and train models
- Discovered statistically significant deviations in trading volume between overnight and daytime sessions, resulting in widening and tightening of spreads and contributing to overall improvement in pricing algorithms

RESEARCH AND PROJECTS

Aarti Singh Group **Carnegie Mellon University**
Graduate Research Assistant *September 2026 - Present*

- Researching whether LLM proactive behaviors drive user reliance misaligned with users' stated long-term goals
- Designing computational benchmarks to evaluate proactivity and alignment and post-train models to optimize for long-term learning goals as opposed to short-term task completion in math, writing and coding domains

EEG Motor Imagery Classification **UCLA**
Github *March 2026*

- Benchmarked 5 neural architectures (EEGNet, TCN, CNN+LSTM, CNN+GRU, Transformer) for cross-subject EEG motor imagery classification on BCI Competition Dataset under subject-dependent and LOSO protocols
- Built training/evaluation framework in PyTorch with Leave-One-Subject-Out cross-validation (36 folds), early stopping, and data augmentation; ran experiments on GCP GPU instances to determine best models
- Conducted frequency band ablation study across 6 physiological bands, finding that mu+beta (8–30 Hz) band captures over 99% of cross-subject discriminative information, consistent with literature.

SKILLS AND INTERESTS

- **Skills and Technologies:** Python (Numpy, Pandas, PyTorch), FastAPI, ML Modeling, A2A Protocol, Git
- **Interests:** Puzzle Games, Chess, GS Warriors, Urban Transit, Michael Lewis Novels, Hiking National Parks